

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 25

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ (1) • $N(N - 1) = 100 \times 99 = 9900$ (1) • $D = 9900 / 2900 = 3.41$ (accept 3.4) (1)	3
01.2	<i>1 mark for each point.</i> • Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B (1) • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
02.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses (1) • Setting lines at night when birds are not feeding (1) • Bird-scaring streamers / decoys (1) • Curved hooks that catch tuna but not birds (1)	3
02.2	<i>1 mark for each point.</i> • Effort: limits on boat size, engine power, days at sea, number of licences; decommissioning payments to reduce fleet size where overfishing is unavoidable (1) • Methods: bans or length limits on drift nets; bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) (1) • Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes) (1)	3

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> - Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) (1) - Juveniles are released when survival chances are good, increasing recruitment to the adult population (1)	2
02.4	<i>1 mark for each point.</i> - Egg / larval survival varies greatly between years so adult numbers do not predict recruitment (1) - Several poor years would lower the MSY; egg surveys allow catches to be reduced in advance rather than after a sudden collapse (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
03.1	<i>1 mark for each point.</i> - Ecological footprint: the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced (1) - Biocapacity: the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares (1)	2
03.2	<i>1 mark for each point.</i> - A hectare of land / sea with the world-average biological productivity (1) - Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared (1)	2
03.3	<i>1 mark for the correct answer.</i> $8.1 / 1.7 = 4.8$ (accept 4.7-4.8) (1)	1
03.4	<i>1 mark for each point.</i> - Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint (1) - Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO₂ accumulates (overshoot) (1) - Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up; it falls earlier each year (8 August in 2016) (1)	3
03.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Renewable energy and energy conservation (carbon footprint is the largest component) (1) - Reduced meat consumption / food waste; local seasonal food (1) - Circular economy: repair, re-use, recycling; longer product lives (1) - Public transport, efficient buildings (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	
04.1	<i>1 mark for each point.</i> - The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) (1) - The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino (1) - Very low success rates with wildlife so far; deep-frozen cells could allow clones even after the last individuals die (1)	3

Question	Marking guidance	Marks
04.2	<i>1 mark for each point.</i> • Micro-propagation: tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant (1) • Black poplar: cuttings / pollarded stems of pure native trees are rooted to produce many identical trees (1) • Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity) (1)	3
04.3	<i>1 mark for each point.</i> • Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild (1) • Seeds collected worldwide; target of 10 000 seeds per species to maintain a diverse gene pool (1) • Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident (1) • Seeds germinated periodically to check viability and replenish stocks; can be used for reintroduction / breeding (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	
05.1	<i>1 mark for each point.</i> • General physiological solvent: most biological reactions take place between substances dissolved in water (1) • Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes (1) • High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) (1) • Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats (1)	4
05.2	<i>1 mark for each point.</i> • From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) (1) • Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ (1) • The ozone layer reduced UV reaching the surface so organisms could survive out of the water (1) • Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out (1)	4
05.3	<i>1 mark for each point.</i> • Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels (1) • This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
06.1	<i>1 mark.</i> • Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1

Question	Marking guidance	Marks
06.2	<i>1 mark for each point.</i> - Impact depends on total population x per capita impact (1) - With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added (1) - Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates (1)	3
06.3	<i>1 mark for each point.</i> - Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage (1) - Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need (1)	2
06.4	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> - Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity (up to 2) - In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) (up to 2) - Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production (up to 2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
07.1	<i>1 mark for each point.</i> - Isolation makes colonisation difficult: only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them (1) - Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists (1) - Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii) (1) - Endemics have small ranges and populations so are vulnerable to extinction (1)	4
07.2	<i>1 mark for each point.</i> - The birds evolved without mammal predators so nest on the ground / in burrows with no defence (1) - Rats eat eggs and chicks; numbers fell to very low levels (45 and 12 pairs) (1) - After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor (1)	3
07.3	<i>1 mark for each point.</i> - Introduced mosquitoes carry avian malaria to which endemic birds have no resistance; birds survive only above about 1500 m where it is too cold for malaria (1) - Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark.</i> Control of developments and visitor numbers (biosecurity, quarantine of imports); protected areas; captive breeding of endemics (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4.	
08.1	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> - Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity (up to 2) - Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes (up to 2) - Judgement depends on species, market and site (up to 2)	4
08.2	<i>1 mark for each point.</i> - Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation (1) - Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow (1) - After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg (1)	3
08.3	<i>1 mark for each point.</i> - About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection (1) - Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose (1) - Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
09.1	<i>1 mark for each point.</i> - Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores (1) - So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats (1)	2
09.2	<i>2 marks for each point (partial credit for a partly correct point).</i> - Competitor: grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites; rhododendron shades ground and releases toxins preventing native regrowth (2) - Predator: American mink (escaped fur farms) reduced water vole range; cane toads in Australia; cats / rats / pigs on islands eating ground-nesting birds; Nile perch in Lake Victoria ate cichlids to extinction (2) - Pathogen: squirrel pox virus carried by grey squirrels kills red squirrels; crayfish plague carried by signal crayfish kills white-clawed crayfish; Dutch elm disease, ash dieback, sudden oak death imported on plants / soil (2)	6

Question	Marking guidance	Marks
09.3	1 mark for each point. - Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids (1) - The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes - Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time - Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY - Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators - Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management - Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
10.2	1 mark. Any taxonomic group of organisms classified together by similarity (eg species, genus, family) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Question	Marking guidance	Marks
11.1	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how human activities influence biodiversity, using examples from a range of different contexts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Direct exploitation: food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil); deliberate eradication of predators, competitors, vectors, pests • Changes in abiotic factors: water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage; inactivity (stopping plagioclimax management) • Changes in biotic factors: introduced competitors, predators, pathogens, hybridisation; loss of pollinators, seed dispersers, keystone species; food chain impacts (sea otters, puffins, jellyfish) • Habitat destruction: deforestation, farmland expansion, urbanisation, mining, reservoirs; fragmentation and edge effects • Contexts: temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica; agriculture, fishing, forestry • Positive influences: habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	25